

- Instructions:
1. It is compulsory to answer all the questions (1.5 marks each) of Part -A in short.
 2. Answer any four questions from Part -B in detail.
 3. Different sub-parts of a question are to be attempted adjacent to each other.

PART -A

Q1 (a) Find the rank of $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$ (1.5)

(b) Show that the radius of curvature at the point $\frac{a}{4}, \frac{a}{4}$ on the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ is $\frac{a}{\sqrt{2}}$. (1.5)

(c) Prove that $B(m, n) = B(n, m)$ (1.5)

(d) Evaluate $\int_0^1 \int_0^{x^2} e^{y/x} dy dx$ (1.5)

(e) Find curl of a vector point function $(x^2 + xy^2)\hat{i} + (y^2 + x^2y)\hat{j}$. (1.5)

(f) Evaluate $\int_1^2 \int_{-\sqrt{2-y}}^{\sqrt{2-y}} 2x^2 y^2 dx dy$ (1.5)

(g) Show that the function $u = x + y - z$, $v = x - y + z$, $w = x^2 + y^2 + z^2 - 2yz$ are not independent of one another. Also find the relation between them. (1.5)

(h) Expand e^{xy} at $(1, 1)$ up to three terms. (1.5)

(i) Find grad ϕ when $\phi = 3x^2y - y^3z^2$ at $(1, -2, -1)$. (1.5)

(j) Examine whether $u(x, y) = \frac{x^2}{y}$, is homogeneous. if yes, find the degree (1.5)

PART -B

Q2 (a) Evaluate $\int_C \vec{F} \cdot d\vec{r}$, where (7)

$\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ and the curve C is rectangle in the xy - plane bounded by $y = 0, x = a, y = b, x = 0$.

(b) Use Green's theorem to evaluate $\oint_C [(y - \sin x)dx + \cos x dy]$, where C is the plane triangle enclosed by the lines $y = 0, x = \frac{\pi}{2}$ and $y = \frac{2}{\pi}x$. (8)

Q3 (a) Evaluate $\int_V (2x + y) dV$, where V is closed region bounded by the cylinder $z = 4 - x^2$ and the planes $x=0, y=0, y=2$ and $z=0$. (7)

(b) Find the area included between the curves $y^2 = 4a(x + a)$ and $y^2 = 4b(b - x)$ (8)

Q4 (a) Solve by rank method the system of equations $x+2y-5z = -9, 3x-y+2z = 5, 2x+3y-z = 3$. (7)

(b) $4x-5y+z = -3$

Find the characteristic equation of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix}$ and show that Λ satisfies the equation. Hence evaluate A^{-1} and A^4 . (8)

Q5 (a) Find the Asymptotes of

$$(y - x)(y - 2x)^2 + (y + 3x)(y - 2x) + 2x + 2y - 1 = 0 \quad (7)$$

(b) Find the radius of curvature of $x = a \cos^3 t, y = a \sin^3 t$ on the curve (8)

$$x^{2/3} + y^{2/3} = a^{2/3}$$

Q6 (a) Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube. (7)

(b) Prove that $\int_0^1 x^\alpha (\log x)^n dx = \frac{(-1)^n n!}{(\alpha+1)^{n+1}}$ (8)

Q7 (a) Verify Euler's Theorem for the functions $f(x, y) = \frac{x^2(x^2 - y^2)^3}{(x^2 + y^2)^3}$ (7)

(b) Investigate the value of λ and μ so that the equations $2x+3y+5z=9, 7x+3y-2z=8, 2x+3y+\lambda z=\mu$ have (i) no solution (ii) unique solution and (iii) infinite number of solutions. (8)
